CO_2 Time Series Analysis and Forecasting

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Executive Summary

Global warming is largely driven by rising CO_2 concentrations, a relationship that is both statistically and scientifically established. This project examines how CO_2 levels correlate with global temperature increases and forecasts when critical thresholds will be exceeded. Using the Keeling Curve (Figure A) and historical CO_2 data from Mauna Loa, we applied time series analysis and ARIMA modeling (Figure H) to predict future trends. Our model, validated through statistical testing (Figures C, D), indicates with 70% certainty that global temperatures will surpass the 2.0°C limit set by the Paris Climate Accord by August 2059 (Figure I). This threshold is significant because exceeding it could trigger severe environmental and economic consequences. By converting forecasted CO_2 levels into expected temperature changes using radiative forcing principles, our projections align with broader scientific warnings about climate change. The results emphasize the urgent need for emissions reduction to mitigate the irreversible impacts of global warming.

1 Introduction and Data

The Keeling Curve, developed in the 1950s by Charles David Keeling, tracks atmospheric CO_2 levels over time (Figure A). Since 1958, data collection has shown a continuous rise in CO_2, which is strongly linked to global warming. This greenhouse gas effect traps heat from the Sun, posing significant risks to life on Earth. Our project models this relationship to predict when global temperatures will exceed the danger threshold set by the Paris Climate Accord (Figure I). To examine the variation of atmospheric CO_2 levels and their impact on average global temperatures over time, we performed a time series analysis using the measurements taken from an observation station atop Mauna Loa[1]. The dataset consists of monthly observations of atmospheric CO_2 concentrations (PPM)[2] from 1958 to 2019 (Figures A, B). To ensure proper data continuity, missing CO_2 values were filled using a cubic spline technique as implemented by the Keeling Institute.

2 CO_2 and Temperature Anomalies

To estimate global temperature anomalies and model climate change, we must first define global average temperature and its connection to CO_2 levels. This value is a mathematical construct rather than a directly observable measurement. We used NASA's GISS Surface Temperature (GISTEMP), which combines land and sea surface temperature anomalies. Scientists gather air and water temperature data worldwide, converting absolute values into anomalies by comparing them to long-term averages. Since these averages can vary, baseline adjustments are necessary. The dataset's reference period is 1951-1980, and NASA indicates that to align with the pre-industrial period (1850-1900), 0.19°C should be added.

[1]The Mauna Loa observatory records CO_2 levels at this more remote and high-altitude location to avoid potential interference from local factors like air pollution, human activities, or natural events like volcano eruptions

[2]The µmol CO_2 per mole of dry air (µmol/mol), which is equivalent to parts per million (ppm). A mole of dry air has the mass of 28.79g with the dry air being the product of factoring out water vapor which dilutes the measurements of CO_2

Thus, for consistency with our CO_2 data starting in February 1959, we used an initial anomaly of 0.07°C, adjusted by 0.19°C, resulting in a starting value of 0.26°C for the global temperature anomaly.

To convert forecasted CO_2 concentrations into temperature anomaly changes, we used a formula derived from Radiative Forcing, which is widely recognized for establishing causation rather than just correlation. CO_2 is considered the dominant driver of global warming (70%), and its impact follows a log() relationship with temperature changes:

$$\Delta T_t = S \log_2\left(\frac{C_t}{C_{t-1}}\right) \tag{1}$$

where Ct is the CO_2 concentration at time t, and Ct-1 is the previous time step. S, the climate sensitivity factor, represents how much global temperature increases when CO_2 levels double. We selected S = 2.37°C, a commonly used value in scientific modeling, which falls within the estimated range between equilibrium climate sensitivity (ECS = 3.0°C) and transient climate response (TCR = 1.7°C). Using this formula, we calculated monthly temperature changes based on variations in CO_2 levels over time.

3 Time Series Modeling and Diagnostics

We analyzed the stationarity and autocorrelation of the CO_2 data using an ACF test, which revealed a gradually decaying autocorrelation pattern (Figure C), indicating a trend and non-stationarity. This was further confirmed by a KPSS test (Figure C), which produced a statistic of 10.38 and a p-value of 0.01, providing strong evidence to reject the null hypothesis of stationarity. Given these traits—non-stationarity, a clear trend, and autocorrelation—we applied differencing and decomposition techniques to properly model the data.

Using the insights gained from our dataset, we applied Seasonal-Trend Decomposition using Loess to separate the CO_2 series into trend, seasonal, and remainder components. We then used a decomposition model to select a suitable ARIMA model for the seasonally adjusted series, which was determined to be ARIMA(1,2,2) (Output A). Thus, a second-order differencing was used to establish a forecastable dataset with stationarity, along with one autoregressive and two moving average terms (AR1 = 0.178, MA1 = -1.5449, MA2 = 0.5525) (Output A). For the seasonal component, a separate seasonal ARIMA(0,0,0)(0,1,0) model captured strong annual seasonality. It is important to note that we used a seasonal window of 23. The 12th and 24th autocorrelation coefficients for the residuals from our ARIMA model were statistically significant. To account for this, we tried different values of the season window, until the ACF of the residuals fell reasonably within the significance bands, indicating proper extraction of pattern information.

Favorable values for sigma-squared (0.07839 and 0.0001514 respectively) and information criteria (AIC = 220.25, AICc = 220.3, BIC = 238.62 for the seasonally adjusted series, AIC = -4288.05, AICc = -4288.04, BIC = -4283.47 for the seasonal model) (Output A) suggest a good overall fit, validating the choice of STL decomposition combined with ARIMA modeling. Diagnostic checks further confirmed model adequacy. Residual time plots showed no visible patterns, the residual ACF was within confidence bounds, and the Anderson-Darling normality test yielded a p-value of 0.5592 (Figure D). Thus, there was no evidence of the residuals deviating from normality or independence per the residual tests we ran. With both the data converted for comparing temperature to CO_2 and a model properly picked for application, we could then forecast in and out the sample.

4 Forecasts and Interpretation

Using the fitted ARIMA models, we generated forecasts of atmospheric CO_2 over 6-year and 20-year horizons, each with a 95% confidence interval. These projections confirm a continued upward trend

with seasonal fluctuations, reinforcing expectations of rising CO_2 concentrations. Applying our CO_2-to-temperature equation, we estimated that global temperatures will reach the critical 2.0°C anomaly above pre-industrial levels by August 2059, aligning with the Paris Climate Accord's warning threshold.

To ensure a smooth transition between observed and forecasted anomalies, we introduced a 5-year blending procedure, gradually shifting weights from historical to predicted values to prevent abrupt jumps. The final blended forecast, visualized alongside historical data and key thresholds (1.5°C and 2.0°C above pre-industrial levels), suggests a steady warming trajectory likely to exceed 1.5°C within the next two decades. Unfortunately, our projections for the 2.0°C threshold align with broader scientific estimates, underscoring the urgency of reducing greenhouse gas emissions. Our model's accuracy was validated by in-sample forecasts, which closely mirrored observed data. The CO_2 versus temperature plot exhibits a clear, increasingly volatile upward trend driven by seasonal variations. Residual plots confirm no discernible trend beyond static noise, indicating robust model performance. The seasonal component primarily reflects the natural cycle of plant respiration, where CO_2 levels decline in spring and summer due to active plant absorption and rise in winter as plants decay and release stored CO_2.

5 Discussion and Implications

While CO_2 is estimated to account for 70% of climate change, additional factors such as methane, nitrous oxide, fluorinated gases, aerosols, weather events, and deforestation contribute to variability. The absence of these secondary drivers may explain some forecasting errors in temperature anomalies. Aerosols, for instance, reflect sunlight and can temporarily counteract warming, while large-scale weather patterns like Southern Oscillations can cause short-term cooling effects. Had these influences been incorporated into our model, the forecasting range would likely be wider to capture the dynamic push-and-pull of multiple climate forces. Overall, while external factors may introduce short-term fluctuations, the dominant trend remains clear: without intervention, global temperatures will continue rising, bringing us closer to irreversible climate consequences.

The Paris Climate Accord, established in 2015, set a global goal to prevent average temperatures from rising beyond 1.5-2.0°C above pre-industrial levels. This threshold is not arbitrary; historical evidence shows that a mere 2.0°C drop in global temperatures triggered the Little Ice Age (14th-19th century), leading to widespread famine and economic hardship. A more drastic 5.0°C drop sparked the last major ice age, making large parts of the planet uninhabitable for thousands of years. By contrast, a rise of 1.5-2.0°C could have equally severe consequences, with mass species extinctions and habitat loss already projected. Many scientists warn that surpassing this limit could push Earth toward its next mass extinction event, showing the need to curb CO_2 emissions.

While this outlook may seem dire, our analysis does not imply that climate change is beyond human control. The planet and humanity still have the means to regulate global temperatures—if decisive action is taken. Shifting away from fossil fuels to renewable energy sources can significantly reduce CO_2 emissions, buying crucial time for ecosystems to recover. Given CO_2's long atmospheric lifespan, immediate reductions in emissions could have a cascading effect in mitigating global warming. Ultimately, our forecast is not a prediction of an inevitable disaster, but a call to action—highlighting what can be prevented rather than what must come to pass.

A Figures and Model Output

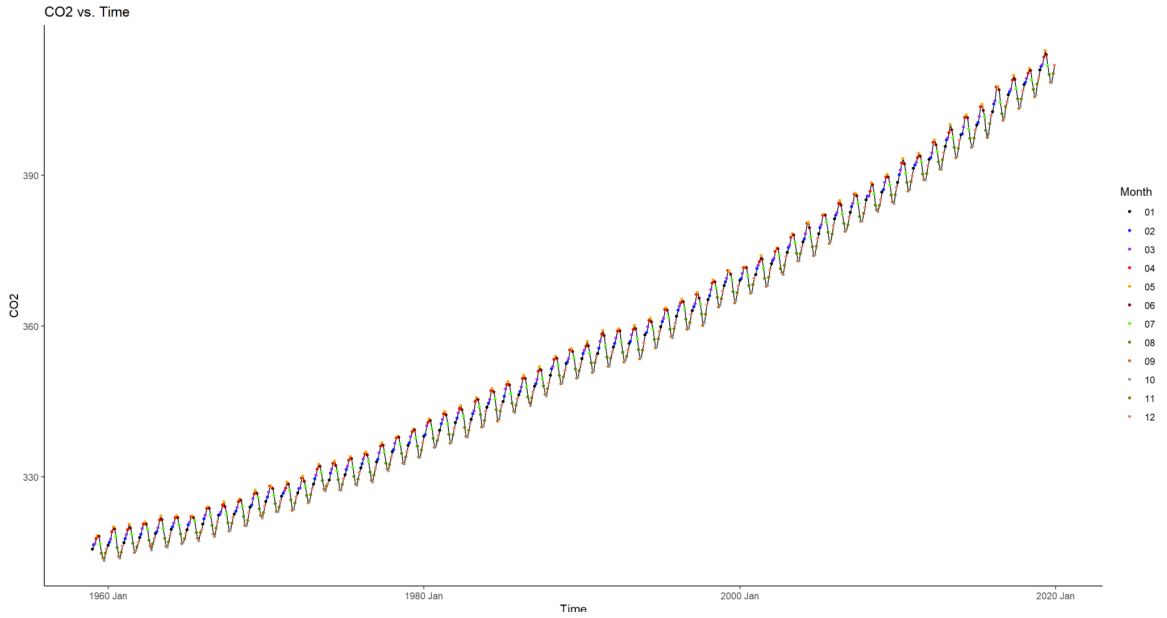


Figure A. Atmospheric CO_2 concentration (ppm) over time.

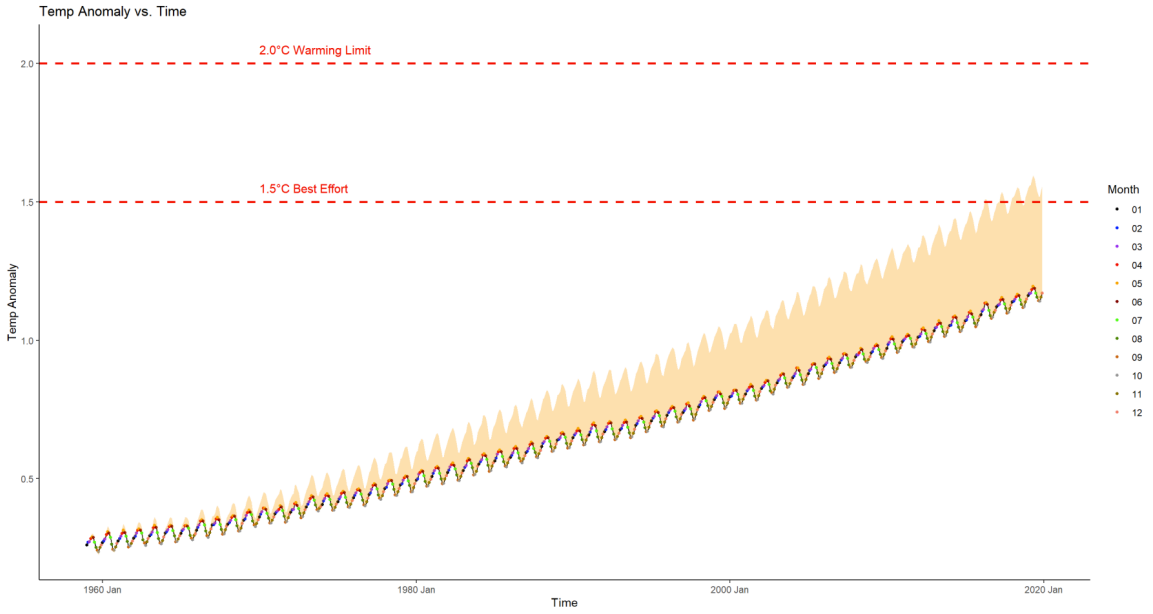
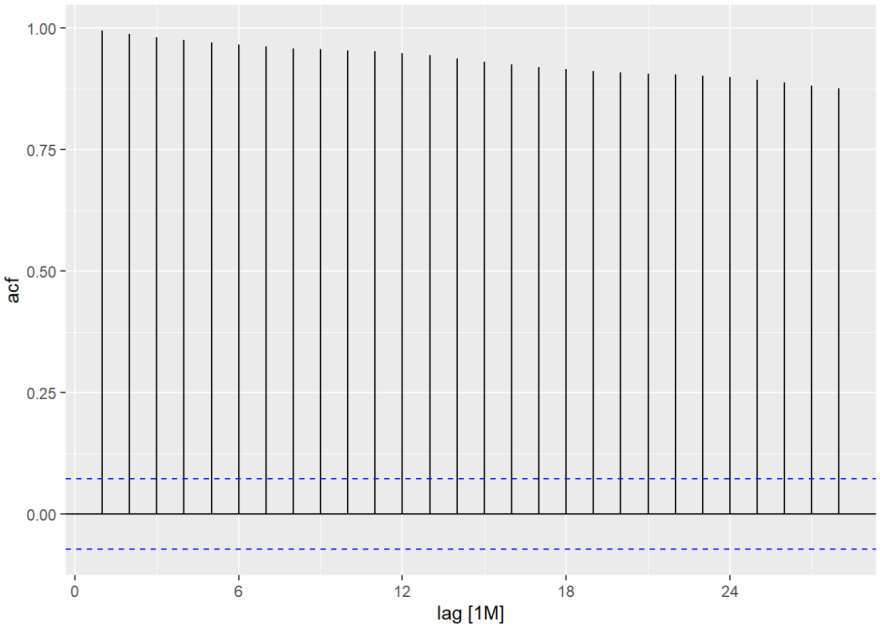
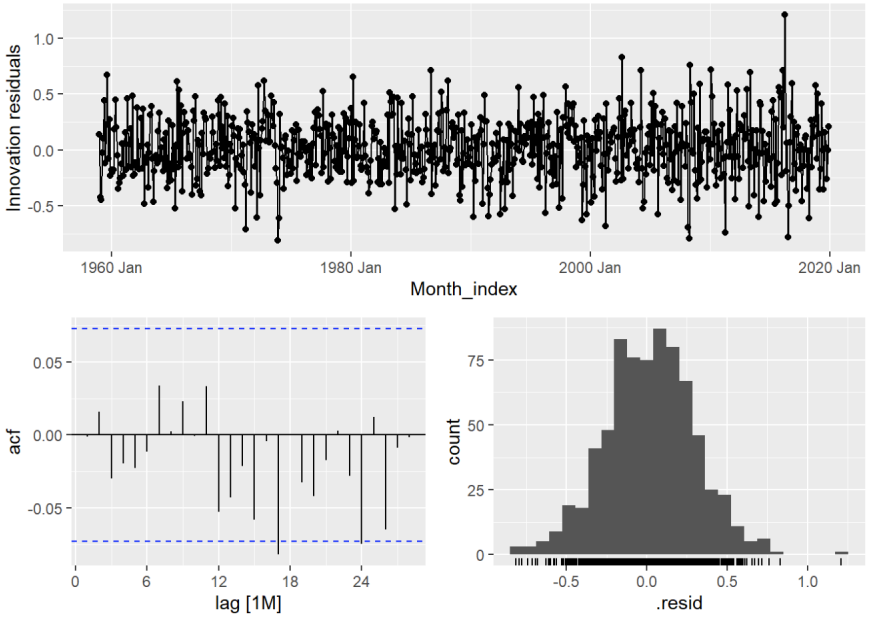


Figure B. In-sample temperature anomaly forecast as a function of atmospheric CO_2 concentration.




```
##   kpss_stat kpss_pvalue
##    10.38394     0.01000
```

Figure C. Autocorrelation and KPSS diagnostics for the non-stationary CO_2 series.




```
##
##  Anderson-Darling normality test
##
## data:  result_ARIMA_augment$.resid
## A = 0.3083, p-value = 0.5592
```

Figure D. Residual diagnostics and Anderson–Darling normality test.

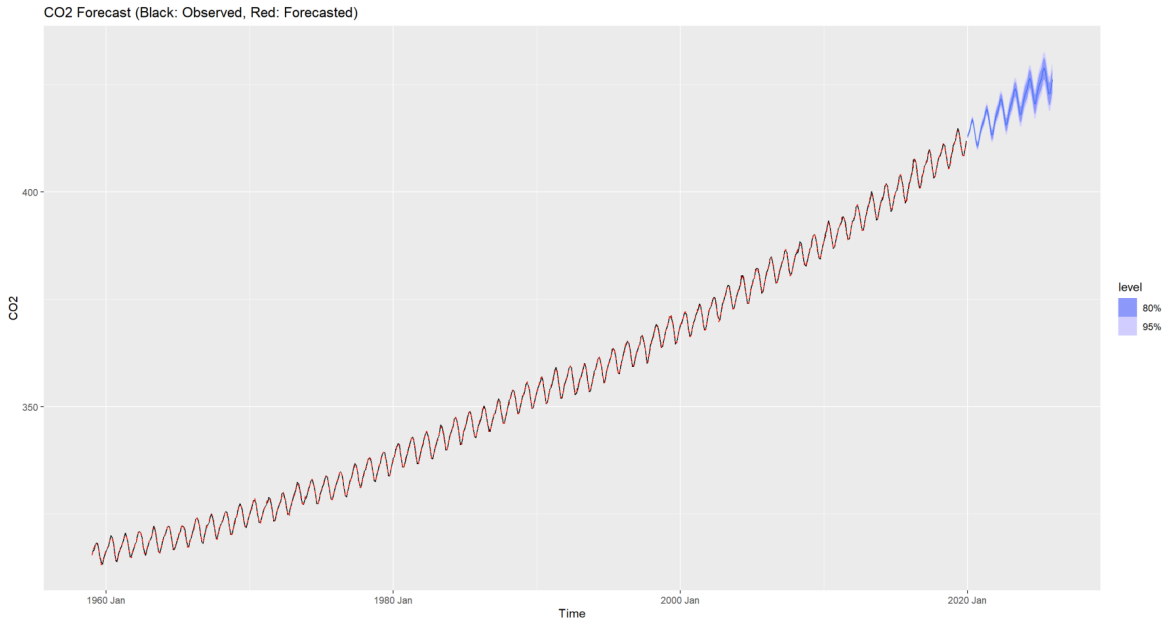


Figure E. Forecast of atmospheric CO_2 concentrations from 2020 through the subsequent years.

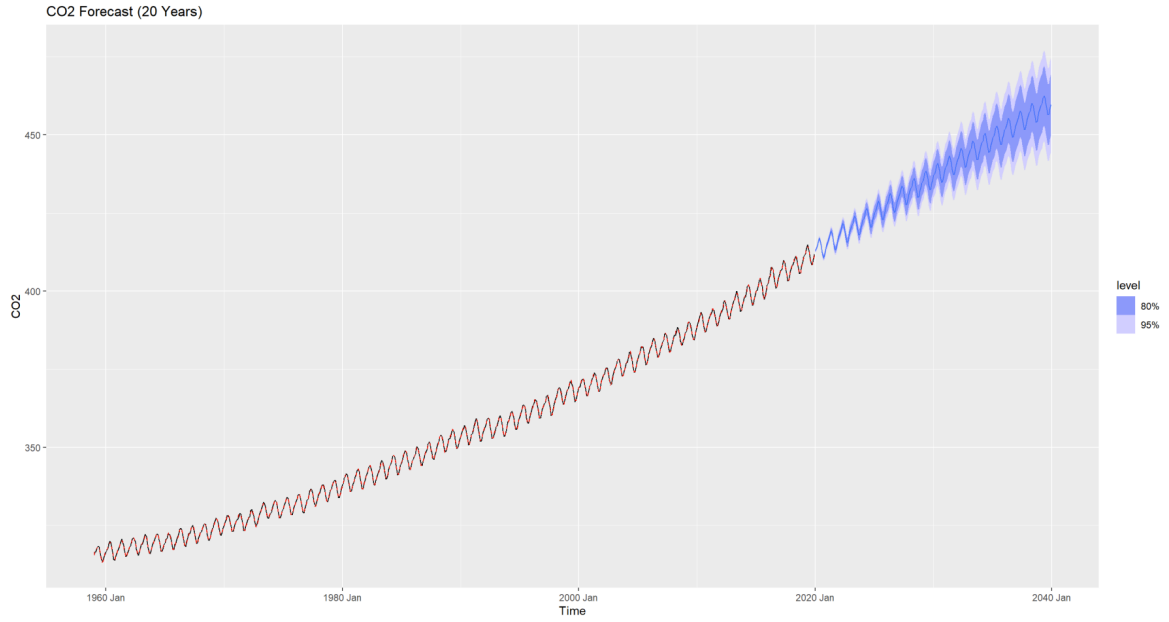


Figure F. Twenty-year forecast of atmospheric CO_2 concentrations to 2040.

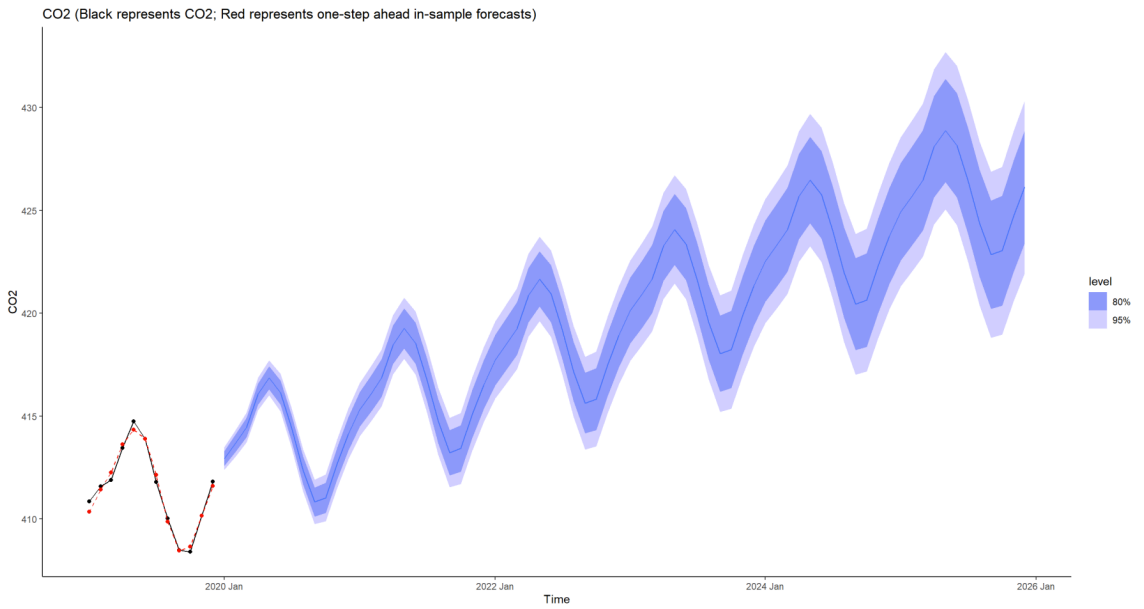


Figure G. One-step-ahead in-sample forecasts compared with observed values.

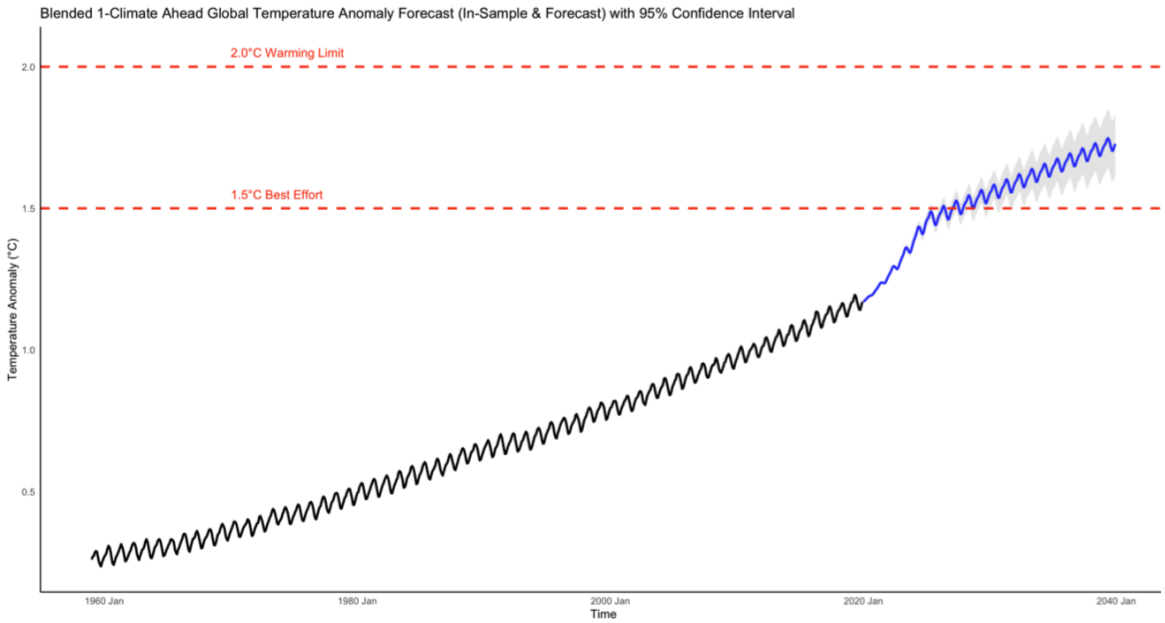


Figure H. Observed, in-sample, and forecast temperature anomalies derived from CO_2 concentrations.

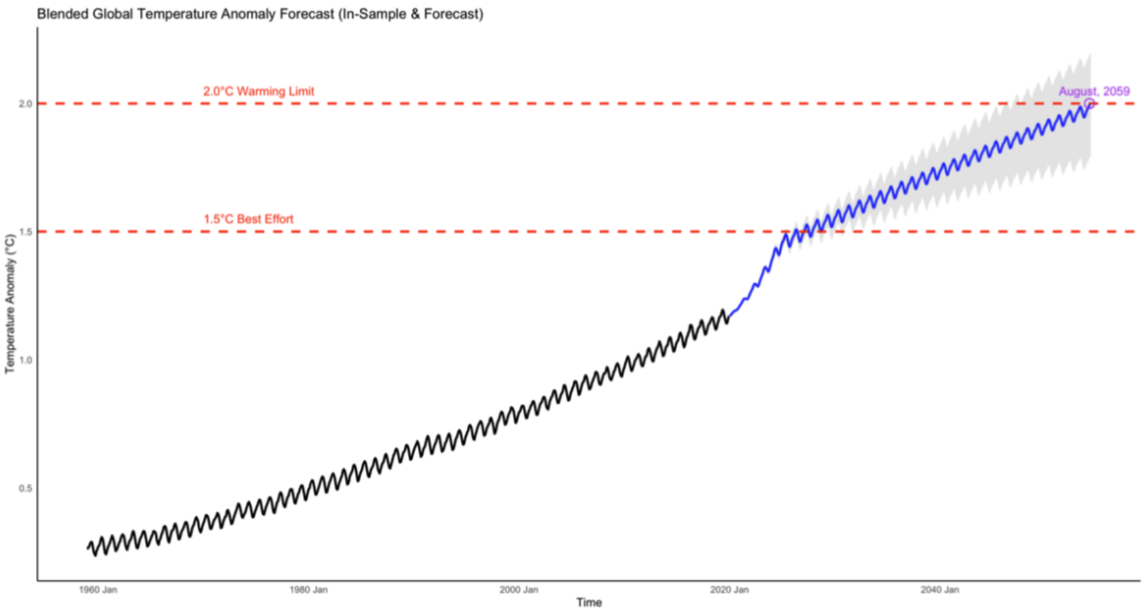


Figure I. Long-horizon temperature anomaly projection and the Paris Agreement reference level.

B Model Output

```
> report(result_dcmp_ARIMA_SARIMA)
Series: CO2
Model: STL decomposition model
Combination: season_adjust + season_year

====================================

Series: season_adjust
Model: ARIMA(1,2,2)

Coefficients:
        ar1      ma1     ma2
      0.178  -1.5449  0.5525
s.e.  0.087   0.0737  0.0730

sigma^2 estimated as 0.07839:  log likelihood=-106.12
AIC=220.25   AICc=220.3   BIC=238.62

Series: season_year
Model: ARIMA(0,0,0)(0,1,0)[12]

sigma^2 estimated as 0.0001514:  log likelihood=2145.03
AIC=-4288.05   AICc=-4288.04   BIC=-4283.47
```

Output A. Selected ARIMA model and information criteria.

```
> hilo(forecast_result_dcmp_ARIMA_SARIMA_today%>% select(-.model))
# A tsibble: 72 x 5 [1M]
   Month_index           CO2 .median                `80%`                  `95%`
         <mth>        <dist>   <dbl>               <hilo>                 <hilo>
 1    2020 Jan N(413, 0.079)    413. [412.5798, 413.2979]80 [412.3897, 413.4880]95
 2    2020 Feb  N(414, 0.11)    414. [413.2728, 414.1226]80 [413.0479, 414.3475]95
 3    2020 Mar  N(414, 0.14)    414. [413.9796, 414.9244]80 [413.7296, 415.1745]95
 4    2020 Apr  N(416, 0.16)    416. [415.5531, 416.5834]80 [415.2804, 416.8561]95
 5    2020 May  N(417, 0.19)    417. [416.3081, 417.4193]80 [416.0140, 417.7134]95
 6    2020 Jun  N(416, 0.22)    416. [415.5475, 416.7363]80 [415.2329, 417.0509]95
 7    2020 Jul  N(414, 0.24)    414. [413.7796, 415.0434]80 [413.4451, 415.3779]95
 8    2020 Aug  N(412, 0.27)    412. [411.6869, 413.0237]80 [411.3331, 413.3776]95
 9    2020 Sep   N(411, 0.3)    411. [410.1245, 411.5326]80 [409.7518, 411.9053]95
10    2020 Oct  N(411, 0.33)    411. [410.2855, 411.7634]80 [409.8943, 412.1546]95
# ℹ 62 more rows
```

Output B. Ten-step-ahead forecast with 80% and 95% prediction intervals.

	Month_index	CO2	CO2_median	CO2_lower	CO2_upper	Time	c0	delta_CO2	d
	<mth>	<dist>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	
1	2053 Dec	N(493, 227)	493.	464.	523.	1140	492.	1.43	
2	2054 Jan	N(495, 228)	495.	465.	524.	1141	493.	1.20	
3	2054 Feb	N(495, 229)	495.	466.	525.	1142	495.	0.772	
4	2054 Mar	N(496, 231)	496.	466.	526.	1143	495.	0.757	
5	2054 Apr	N(498, 232)	498.	468.	528.	1144	496.	1.62	
6	2054 May	N(499, 234)	499.	469.	529.	1145	498.	0.796	

delta_Temp	Forecasted_Avg_Global_Temp_Anomaly	c0_lower	delta_CO2_lower	delta_Temp_lower	Forecasted_Avg_Global_Temp_Anomaly_Lower
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
0.00989	1.97	463.	1.33	0.00982	1.76
0.00833	1.98	464.	1.11	0.00816	1.77
0.00533	1.98	465.	0.677	0.00497	1.78
0.00522	1.99	466.	0.661	0.00485	1.78
0.0111	2.00	466.	1.52	0.0111	1.79
0.00546	2.00	468.	0.700	0.00511	1.80

c0_upper
<dbl>
521.
523.
524.
525.
526.
528.

	delta_CO2_upper	delta_Temp_upper	Forecasted_Avg_Global_Temp_Anomaly_Upper
	<dbl>	<dbl>	<dbl>
1	1.52	0.00996	2.16
2	1.30	0.00848	2.17
3	0.867	0.00565	2.18
4	0.852	0.00554	2.18
5	1.71	0.0111	2.19
6	0.891	0.00577	2.20

Output C. Tail of the extended forecast and 95% intervals for CO_2 concentrations and temperature anomaly.